

Connecting the Dots in AMR Surveillance

How bacteria climb the resistance ladder, and how to forecast the next rung

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Abstract

A pathogen defeats an antibiotic only if it carries the right machinery: an enzyme that destroys the drug, a modified target, a permeability barrier, an efflux pump. Because one mechanism usually covers several drugs, and because mechanisms accumulate rather than replace one another, resistance is not acquired at random. Drugs fall in a broadly consistent order, those a pathogen can defeat with little equipment first and those requiring a large accumulated repertoire last, so what a species already defeats constrains what it will defeat next. Routine surveillance, read one pathogen–antibiotic pair at a time, cannot see that order. We recover it from the Pfizer ATLAS programme distributed through the Vivli AMR Register (394 species, 43 antibiotics, 2004–2024). For every pair we ask a comparative rather than an absolute question: is this species more resistant to this drug than expected, given how resistant it is across all drugs and how often that drug fails across all species? A general rise in resistance therefore produces no signal; a signal appears only when one species pulls away from the field on one drug.

Connecting each species to the drugs it comparatively defeats gives a bipartite network, and that network turns out to be strongly *nested* within coherent groups of organisms and drugs (in-block nestedness 0.386 against 0.145 ± 0.023 for degree-matched randomisations, $z = 10.5$): the drugs defeated by a narrowly resistant species are largely a subset of those defeated by a broadly resistant one. This is the ladder that ordered accumulation predicts, and it is what makes ranking possible. Fitness–Complexity converts it into two scores computed together: the *fitness* of a species measures how much resistance capability it has accumulated, and the *complexity* of a drug measures how much a pathogen must have accumulated before it can defeat that drug: complexity is high only when *every* species defeating the drug is itself well equipped. Both recover knowledge withheld from the calculation: complexity rises across the WHO Access, Watch and Reserve tiers ($\rho = 0.59$, $p < 10^{-4}$), and ESKAPEE taxa sit at the top of their ladders ($p < 10^{-5}$).

The ladder also anticipates the next rung. Two antibiotics are similar when the same species defeat both, which is direct evidence that they require similar mechanisms; scoring each species by the share of a drug’s similarity neighbourhood it already defeats gives a forecast with no fitted parameters. Using only data available at the time, it reached AUC 0.71 for the following year, the top sixth of the ranking crossed the threshold 3.6 times more often than the bottom sixth (16.6% versus 4.6%), and a network frozen in 2019 held AUC 0.68–0.74 over five years. The framework is delivered as an interactive atlas (<https://giordano-demarzo.github.io/resistome-atlas/>) showing, for any pathogen, where it stands and which drugs it is closest to defeating.

Objectives

We pursued three linked objectives: (i) turn routine counts into a pathogen–antibiotic network that isolates *specific* resistance affinities from the general resistance level of a species and the general vulnerability of a drug; (ii) test whether resistance profiles are nested, as the ordered accumulation of mechanisms predicts, and use Fitness–Complexity to place every species and every drug on the resulting ladder (how much capability a species has assembled, how much capability a drug demands); and (iii) test prospectively whether a species’ position relative to *similar* drugs anticipates which drug it defeats next.

Methods

From surveillance counts to a pathogen–antibiotic network

The input contains annual species–antibiotic counts from the Pfizer ATLAS surveillance programme distributed through the Vivli AMR Register, spanning 394 species, 43 antibiotics and 2004–2024 [1]. For each analysis window, resistant and tested counts were pooled before applying the minimum-isolate threshold. A pair (p, a) was retained when at least $n_{\min} = 30$ isolates had been tested, and its resistance fraction was

$$r_{pa} = \frac{n_{pa}^R}{N_{pa}}$$

where intermediate isolates contribute to N_{pa} but not to n_{pa}^R . If the pair was absent or below threshold, r_{pa} was undefined. The workflow was repeated at $n_{\min} = 10$ and 50, all else unchanged.

Raw percentages cannot be compared across such a table. A resistance fraction of 40% may mean that the species resists almost everything, that the drug is failing against everything, or that this species has acquired something specific against this drug. Only the third carries information about mechanisms. We therefore divide the observed fraction by the fraction expected from the two margins alone. With $\mathcal{O}$ the sufficiently observed cells, $R_p = \sum_{a:(p,a) \in \mathcal{O}} r_{pa}$ the overall resistance level

of species p , $A_a = \sum_{p:(p,a) \in \mathcal{O}} r_{pa}$ the overall failure level of drug a , and $T = \sum_{(p,a) \in \mathcal{O}} r_{pa}$,

$$\text{RCA}_{pa} = \frac{r_{pa}}{R_p A_a / T} = \frac{r_{pa} T}{R_p A_a}$$

so $\text{RCA}_{pa} = 2$ means twice the resistance the margins alone would imply. The construction is revealed comparative advantage, introduced to compare economic profiles of unequal size and used here for the same reason [2, 3]. Two properties matter clinically. It is invariant to uniform escalation: if resistance rises proportionally everywhere, every RCA is unchanged, so the analysis tracks the emergence of *specific* mechanisms rather than the general trend a prevalence curve already reports. And a high value can occur at a modest absolute percentage, which is precisely when an emerging mechanism is still actionable.

We call $\text{RCA}_{pa} \geq 1$ a *comparative-resistance link* and write $M_{pa} = \mathbf{1}\{\text{RCA}_{pa} \geq 1\}$. This defines a bipartite network with species on one side and antibiotics on the other, connected wherever the species is comparatively resistant to that drug; in the pooled 2015–2024 window it has 139 species, 40 drugs and 991 links. Everything that follows is read off this single object. Undefined RCA evaluates as false, so $M_{pa} = 0$ merges observed pairs with $\text{RCA}_{pa} < 1$ and insufficiently observed pairs: a zero means “no comparative-resistance link in the analysed universe”, not confirmed susceptibility. RCA is a within-dataset comparative signal, not a clinical breakpoint.

Nested profiles: the resistance ladder

If mechanisms accumulate in a broadly consistent order, the bipartite network must have a particular shape. A species holding few mechanisms defeats few drugs, and those drugs are among the ones defeated by a species holding many: narrow profiles sit inside broad ones, and rare drugs are reached only from the top. Sorting species and drugs by breadth of profile then makes the matrix approximately triangular. This property is called *nestedness*. It is a recurrent feature of bipartite ecological networks: plant-pollinator and other mutualistic communities, host-parasite associations and food webs are typically nested, with the partners of specialists forming subsets of the partners of generalists [4, 5, 6]. A nested resistance matrix likewise indicates that resistance profiles are not arbitrary sets of drugs but successive stages of one accumulation sequence.

Nesting can only hold among species facing the same drugs. A carbapenemase is irrelevant to a staphylococcus and vancomycin is irrelevant to *E. coli*, so a Gram-positive and a Gram-negative organism are not two rungs of one climb, and forcing them onto a single axis destroys the ordering. We therefore partition the network before measuring the ladder. Spectral co-clustering split species and drugs simultaneously into five matched groups [9], using M alone: no taxonomic, drug-class, AWARe or ESKAPEE information entered this step. The partition is a prerequisite for measurement, not a result in itself; what matters is whether each group is nested.

For species i, j with degrees k_i, k_j (the number of drugs each defeats), let $O_{ij} = \sum_a M_{ia} M_{ja}$ be the number of drugs both defeat, with degree-based expectation $\langle O_{ij} \rangle = k_i k_j / A$ over A drug columns. The statistic compares the overlap of a narrower profile with a broader one against that expectation, and analogously for drugs [7, 8]:

$$\mathcal{I}_R = \sum_{\substack{i,j \\ k_i > k_j, b_i = b_j}} \frac{O_{ij} - k_i k_j / A}{k_j (C_i - 1)} \quad , \quad \mathcal{I}_C = \sum_{\substack{a,c \\ k_a > k_c, b_a = b_c}} \frac{O_{ac} - k_a k_c / P}{k_c (C_a - 1)} \quad , \quad \mathcal{I} = \frac{2}{P + A} (\mathcal{I}_R + \mathcal{I}_C)$$

where b identifies a group and C its size. Setting all nodes to one group gives global nestedness $\mathcal{N}$, which measures how well a single ladder describes the entire table; the same distinction separates firm-level from country-level economic networks [8]. Significance was assessed with 100 Bernoulli matrices satisfying $\Pr(M_{pa}^{\text{null}} = 1) = \min(k_p k_a / E, 1)$, $E = \sum_{pa} M_{pa}$, preserving degree heterogeneity in expectation; empty nodes were removed and every null matrix re-clustered, so the null could reproduce group structure by chance.

Fitness and complexity: what a species has accumulated, what a drug requires

Fitness–Complexity scores a species and a drug in one coupled calculation, since the two are defined through each other [3]. A species’ *fitness* rises with the number of drugs it defeats, each weighted by the drug’s *complexity*: defeating a drug that nearly everything defeats says little, defeating one that almost nothing defeats says a great deal. Complexity, in turn, is the accumulated capability a drug requires before it can be defeated, and the calculation infers it from who has managed: complexity is high only if *every* species defeating the drug is well equipped, because the update is harmonic in fitness and a single poorly equipped conqueror collapses the score. This asymmetry is deliberate, since one modestly equipped species defeating a drug is proof that the drug did not require much, and it separates complexity from mere rarity of resistance. Starting from $F_p^{(0)} = Q_a^{(0)} = 1$, the pipeline iterates

$$\tilde{F}_p^{(n)} = \sum_a M_{pa} Q_a^{(n-1)} \quad , \quad \tilde{Q}_a^{(n)} = \left(\sum_p \frac{M_{pa}}{F_p^{(n-1)}} \right)^{-1}$$

normalising F and Q by their means at each step. The iteration converges to an informative ranking only on a nested substrate, which is why the partition matters: scores were computed separately within each group and standardised before

pooling, so they describe position on a local ladder rather than an absolute scale shared across groups. Fitness is a network score, unrelated to evolutionary fitness; complexity is not a measure of drug potency or clinical efficacy.

Two independent references were used only *after* the scores were computed, as external checks: the WHO Access, Watch and Reserve stewardship tiers [10] for antibiotic complexity, and the ESKAPEE list (the ESKAPE group [11] extended here with *Escherichia coli*) for pathogen fitness. Neither entered the network, the grouping or the scores.

The antibiotic similarity network and prospective forecasting

If two antibiotics are repeatedly defeated by the same species, they plausibly require overlapping mechanisms, an empirical similarity established from resistance behaviour rather than from chemical class. Projecting the bipartite network onto its antibiotic side makes this explicit: two drugs are linked in proportion to the number of species that defeat both,

$$W_{ab}^A = \sum_p M_{pa}^{(t)} M_{pb}^{(t)}$$

giving a weighted similarity network over the 40 antibiotics. A species that already defeats a drug’s closest neighbours has, by construction, most of what that drug requires. For each origin year $t = 2019, \dots, 2023$ all quantities were rebuilt using only data from 2004 through t , and for every sufficiently observed pair (p, a) with no link at t we scored the exposure of species p to drug a as the weighted share of a ’s neighbourhood that p already defeats,

$$\rho_{pa}^A = \frac{\sum_o W_{ao}^A M_{po}^{(t)}}{\sum_o W_{ao}^A}$$

The outcome was whether the same sufficiently tested pair had $\text{RCA}_{pa} \geq 1$ in year $t + 1$. The score contains no fitted coefficient, no training step and no tuned threshold: it is read directly off the network. Discrimination was measured by AUC against a baseline given by the drug’s current fraction of linked species (the naive “how common is resistance to this drug” ranking), with 200 node-label permutations for significance. To test persistence, we froze network, scores and candidate set at $t_0 = 2019$ and evaluated them, unchanged, against each target year from 2020 to 2024, restricting evaluation to pairs sufficiently tested in that year.

Results

Resistance profiles are nested: each group is a ladder

Pooling 2015–2024 yielded 1,932 sufficiently observed cells in a 139-by-40 network, of which 991 carried a comparative-resistance link. Within the five groups the nested shape is pronounced: $\mathcal{I}^* = 0.3855$ against 0.1453 ± 0.0229 for degree-matched randomisations ($z = 10.51$). Narrow profiles sit inside broad ones, and the drugs at the top of each group are defeated only by the species that defeat almost everything else in it: resistance within a group is acquired in an order, not as an arbitrary collection (Figure 1a).

Applied to the table as a whole the same statistic is low and *below* the null ($\mathcal{N} = 0.0496$ against 0.0844 ± 0.0081 , $z = -4.30$), a ratio $\mathcal{I}^*/\mathcal{N} = 7.77$ (Figure 1b). This is the expected consequence of comparing organisms that do not face the same drugs, and it carries a practical corollary: there is no single axis on which a *Klebsiella* and an *Enterococcus* can be ordered, so species must be compared with their peers. As a check on the partition, the five groups, inferred from resistance data alone, reproduce organism–drug divisions a clinical microbiologist would recognise: one large Gram-negative group pairing Enterobacterales and non-fermenters with the β -lactams, carbapenems, fluoroquinolones and aminoglycosides; the staphylococci with the macrolides, oxacillin and linezolid; the enterococci with the glycopeptides and daptomycin; and two anaerobe groups (the *Clostridia* with metronidazole and penicillin, the *Bacteroides* with cefoxitin, clindamycin and tigecycline).

Fitness and complexity place species and drugs on the ladder

Within each group the equations converge to a stable ordering: a fitness for every species and a complexity for every drug. These orderings agree with two references withheld from every step of the construction. Antibiotic complexity increases across the WHO AWaRe tiers (Spearman $\rho = 0.5895$, $p = 9.84 \times 10^{-5}$, 38 drugs; $\rho = 0.6244$, $p = 8.49 \times 10^{-4}$ in the largest group), recovering the Access, Watch and Reserve gradient from resistance patterns alone (Figure 2a). Taxa on the ESKAPEE list occupy the upper rungs: median standardised fitness 1.044 for the 16 matching taxa against -0.387 for 107 others (one-sided Mann–Whitney $p = 4.60 \times 10^{-6}$, Figure 2b). The drugs that stewardship policy already protects are the ones the calculation finds most complex, and the pathogens clinicians already regard as hardest to treat are the ones it finds to have accumulated most. Fitness and complexity see only the topology of comparative-resistance links, yet the agreement shows the rankings carry recognisable clinical meaning.

Resistance spreads to similar drugs, and the spread can be anticipated

Projecting the bipartite network onto its antibiotic side gives the antibiotic similarity network (Figure 3a): each node is a drug, two drugs are joined in proportion to the species that defeat both, and mechanistically related agents fall into distinct regions. Highlighting the drugs a single pathogen comparatively defeats, and advancing year by year, shows that resistance does not

appear at scattered points: for *Klebsiella pneumoniae* it ignites in one region and expands outward through it (12 drugs by 2014, 17 by 2019, 21 by 2024), while distant parts of the network stay dark throughout (Figure 3b–d). This is the ladder seen in motion, since each newly defeated drug is a short step from drugs already defeated, and it is the basis of the forecast.

The rolling-origin back-test contained 1,839 eligible pathogen–drug pair-years, of which 131 (7.12%) acquired a link in the following year. Exposure on the antibiotic similarity network, built each year from the raw co-occurrence weights of the binarised network using only data available at the time, reached a pooled AUC of 0.7076 (origin-specific 0.6688–0.7703) against 0.5123, essentially random, for the naive prevalence baseline (Figure 4b). A more demanding baseline ranks each pair by how close its current comparative resistance already sits to the threshold: this RCA-proximity score alone reaches 0.6835, so much of the near-term signal is simply that pairs already near $RCA = 1$ tend to cross it. Network exposure still improves on it (0.7076 versus 0.6835), and the two are only weakly correlated (Pearson 0.38), so the neighbourhood carries information beyond “already nearly resistant” (Figure 4d). An AUC of 0.7076 means a randomly chosen future event outranked a randomly chosen non-event 70.76% of the time, a ranking quality, not an individual probability. The ranking also separates risk in absolute terms, which is what a watch-list requires: in the top sixth of exposure, 51 of 307 pairs (16.61%) crossed the threshold within a year against 14 of 307 (4.56%) in the bottom sixth, a 3.6-fold difference (Figure 4a). The network exposure exceeded all 200 node-label permutations ($p < 0.005$). Per-drug performance was highest for ceftazidime, gentamicin, meropenem, cefepime and ceftriaxone, reaching AUC 0.89 for ceftazidime (Figure 4c); these are the contagion hubs whose neighbourhood most reliably announces the next failure, though each estimate rests on 5–9 events and remains exploratory.

The signal is not a one-year artefact. With network, scores and candidate set frozen at 2019, AUC stayed between 0.681 and 0.742 at horizons of one to five years, while the prevalence baseline fluctuated around chance (0.41–0.58). A network built once therefore remains informative for years. Conclusions were robust to the isolate threshold: repeating the entire pipeline at $n_{\min} = 10, 30, 50$ preserved both the in-group ladder and the ESKAPEE separation, with the default $n_{\min} = 30$ giving the strongest AWaRe association ($\rho = 0.590$ versus 0.547 and 0.442) and the highest back-test AUC (0.708 versus 0.638 and 0.702).

Impact of the work

What this adds to surveillance practice. A surveillance report answers “how much resistance is there?”, not “where is it going, and which pathogen–drug pairs deserve attention before they become a problem?”. This framework answers the second question from the same counts, with no additional data collection. It contributes three things a percentage table cannot: a *position* for every species and drug on the ladder of its own group, identifying both the organisms that have accumulated most and the drugs only they can defeat; a *similarity network* over antibiotics built from resistance behaviour rather than chemical class, which shows which drugs stand or fall together; and a *ranked watch-list* of pairs that are currently unremarkable but sit next to resistance that already exists. Because the network also names the contagion hubs, it indicates where limited testing capacity yields the most early warning, and which drugs belong in a sentinel panel even when resistance to them is currently low.

Interpretability is a design requirement, not a claim. The chain is short and fully inspectable (counts $\rightarrow r_{pa} \rightarrow RCA_{pa} \rightarrow M \rightarrow$ similarity network $\rightarrow$ exposure score), and no classifier is trained, no coefficient fitted, no threshold tuned. A pair appears high on the watch-list because specific, nameable drugs in its neighbourhood are already defeated by that species, and a reviewer can check that reasoning against the isolate counts in minutes. The output can therefore be argued about, and rejected, by a microbiologist rather than accepted on trust, and recomputed the moment new counts arrive.

The Resistome Atlas. The framework is delivered as an interactive dashboard, *The Resistome Atlas* (<https://giordano-demarzo.github.io/resistome-atlas/>), together with a deterministic end-to-end analysis script and the output tables. A user selects a pathogen and watches its resistance ignite across the antibiotic similarity network year by year, inspects where each drug and species sits on its local ladder alongside AWaRe labels, follows resistance timelines, and reads a ranked list of the drugs that pathogen is closest to defeating. The intended uses are practical: a national or hospital surveillance team can see which of its organisms are climbing, a stewardship committee can see which protected drugs sit closest to organisms already resistant to their neighbours, and a laboratory can see which pairs are worth confirmatory testing. The forward probabilities shown in the dashboard come from a separate one-variable calibration of the exposure score, not from the parameter-free evaluation reported here; no output is clinical guidance.

Deployment beyond this dataset. The pipeline needs only isolate counts by species, drug and year. Fractions, margins, groups and networks must be recomputed inside each selected country, region or time window, since comparative resistance is defined relative to the population analysed, which is what makes local re-estimation informative, as the ladder of one hospital network need not match the global one. Temporal re-estimation is demonstrated here; geographical re-estimation is a deployment opportunity rather than a reported result, because the table used here is pooled across countries. The machinery would extend naturally to genomic data, where the mechanisms that the ladder currently implies could be observed directly.

Limits. Comparative resistance is relative: a pair can be flagged at a modest percentage, or be unremarkable yet clinically severe. ATLAS sampling is not population-random, and a zero in M merges unremarkable pairs with untested ones, so the network partly reflects testing coverage. The number of groups is fixed at five, the null is degree-corrected only in expectation, and AWaRe and ESKAPEE give external consistency, not clinical validation. AUC measures ranking quality, not individual risk; a change in comparative status may reflect shifting margins or surveillance composition as well as biology; and nothing

here establishes causation, transmission or treatment failure.

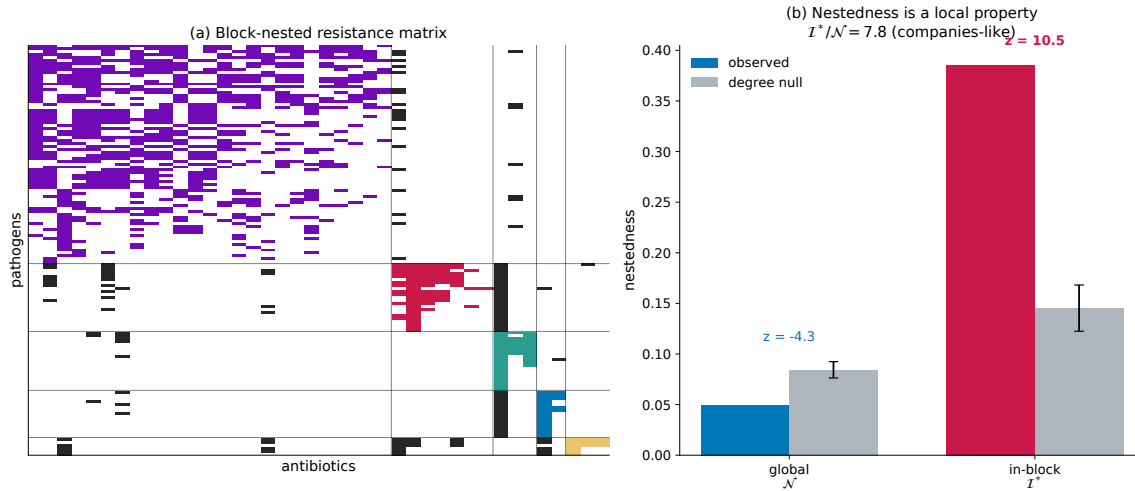


Figure 1: Resistance profiles are nested within groups. In (a), species (rows) and drugs (columns) are ordered by the five groups recovered from the resistance data alone and by breadth of profile within each group; coloured cells are comparative-resistance links inside a matched group, dark cells link different groups. Each group shows the triangular shape expected when species differ in how many resistance mechanisms they have accumulated: narrow profiles sit inside broad ones. Panel (b) compares global nestedness N with in-group nestedness I^* against a degree-matched Bernoulli null; a single ladder spanning all organisms performs *below* chance, local ladders far above it.

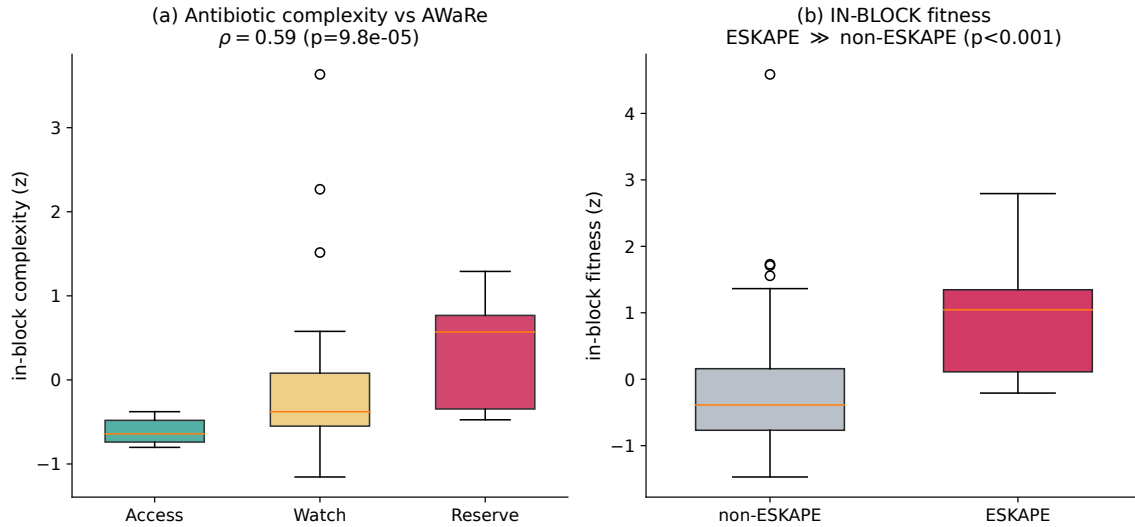


Figure 2: Ladder position recovers withheld biology. (a) Within-group antibiotic complexity, the accumulated capability a drug requires before it can be defeated, rises across the WHO AWARe stewardship tiers, recovering the Access, Watch and Reserve gradient from resistance patterns alone. (b) Within groups, ESKAPEE taxa occupy the upper rungs of their ladders. Neither AWARe nor ESKAPEE was used to build the network, define the groups or compute the scores. Fitness is a network score, not evolutionary fitness.

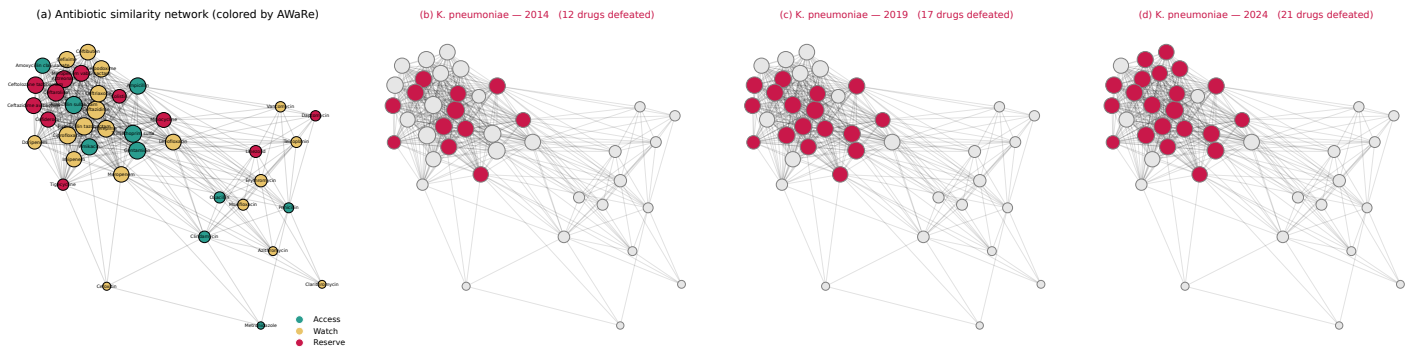


Figure 3: The antibiotic similarity network, and resistance spreading across it. (a) The projected antibiotic similarity network: each node is a drug, two drugs are joined in proportion to the number of species that defeat both, and the force layout groups mechanistically related drugs; colour indicates AWARe tier. (b–d) Drugs that *Klebsiella pneumoniae* comparatively defeats, highlighted on the same layout at three points in time. New nodes light up next to nodes already lit, so resistance expands outward from an occupied region (12 drugs in 2014 to 21 in 2024) while distant parts of the network stay dark, the pattern that makes neighbourhood exposure predictive in Figure 4. Highlighting is cumulative; degree normalisation affects only the layout.

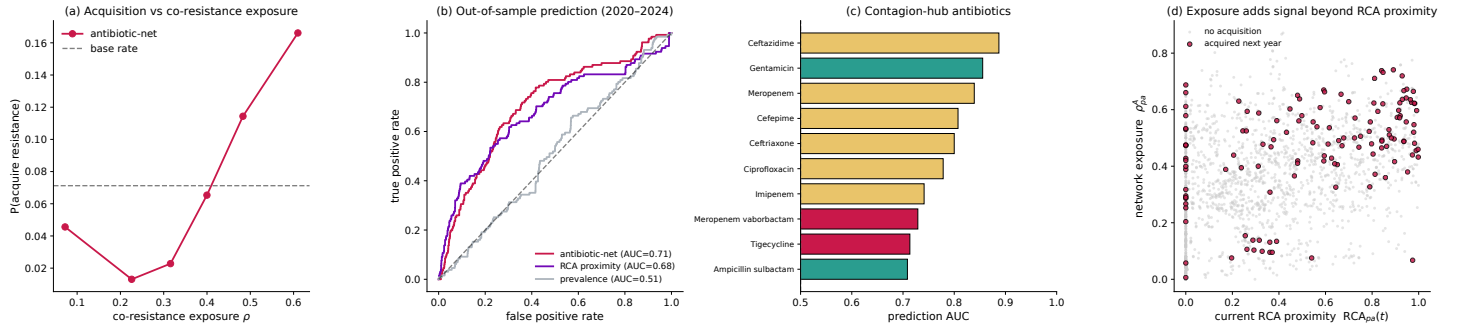


Figure 4: Forecasting the next drug. (a) The probability that a species acquires a comparative-resistance link within one year rises with the share of the drug’s neighbourhood it already defeats. (b) Pooled prospective ROC curves, using only data available before each forecast: network exposure (AUC 0.71) against a trivial baseline that ranks pairs by how close their current comparative resistance already sits to the threshold (RCA proximity, 0.68) and against drug prevalence (0.51). (c) Contagion hubs: the drugs whose neighbourhood most reliably announces the next failure. (d) Network exposure versus current RCA proximity for every candidate pair; pairs that acquire resistance the next year (red) reach high exposure even at low proximity, so the network carries signal beyond “already nearly resistant”. Prediction uses the raw co-occurrence weights of the binarised network. The event is the appearance of a comparative-resistance link, not proof of biological acquisition.

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